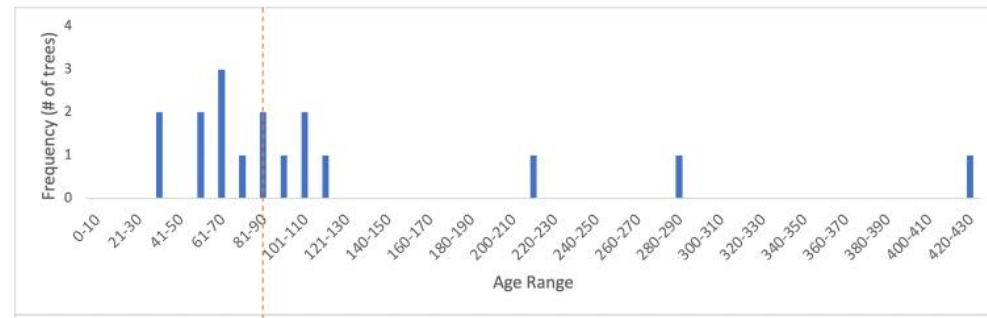
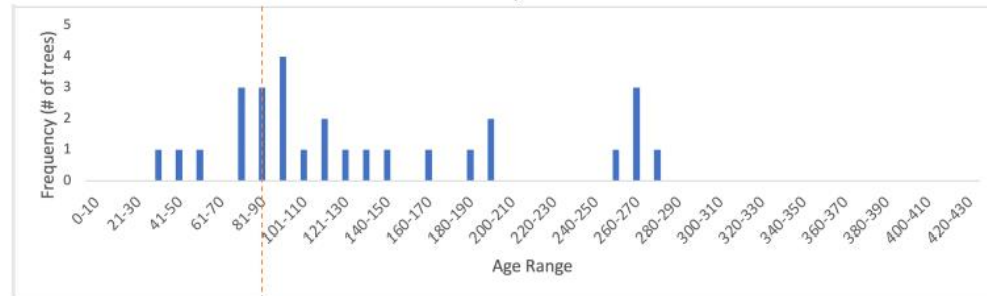




a)



b)

Figure 13: Age range of sampled trees for a) Douglas Fir and b) Engelmann Spruce growing on a limestone cliff at Wasootch Creek as of 2021. Dashed orange line represents the most recent fire (1936). Multiple trees are older than the most recent fire in the area.

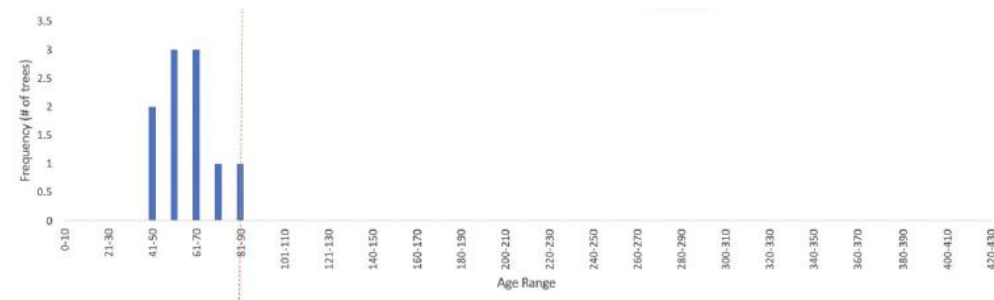


Figure 14: Age range of sampled Douglas Fir and Engelmann Spruce trees growing at glacial till site at Wasootch Creek as of 2021. Dashed orange line represents the most recent fire (1936). Most sampled trees are younger than the most recent fire in the area.

Ring Width and Sapwood Area

The average ring width of cliff trees is significantly smaller than the average ring width of trees of the same species growing at the glacial till site (Figure 15) (Douglas Fir: two sample t-test, $t = 6.0399$, $df = 21$, $p\text{-value} = 5.4e-06$; Engelmann Spruce: two sample t-test, $t = -6.0265$, $df = 34$, $p\text{-value} = 7.956e-07$).

Trees of both species growing on the cliff have significantly smaller sapwood area than trees of the same species growing at the glacial till site (Figure 16) (Engelmann Spruce: Kruskal Wallis Pairwise Comparison, $p=0.00069$; Douglas Fir: Kruskal Wallis Pairwise Comparison, $p=0.00073$).

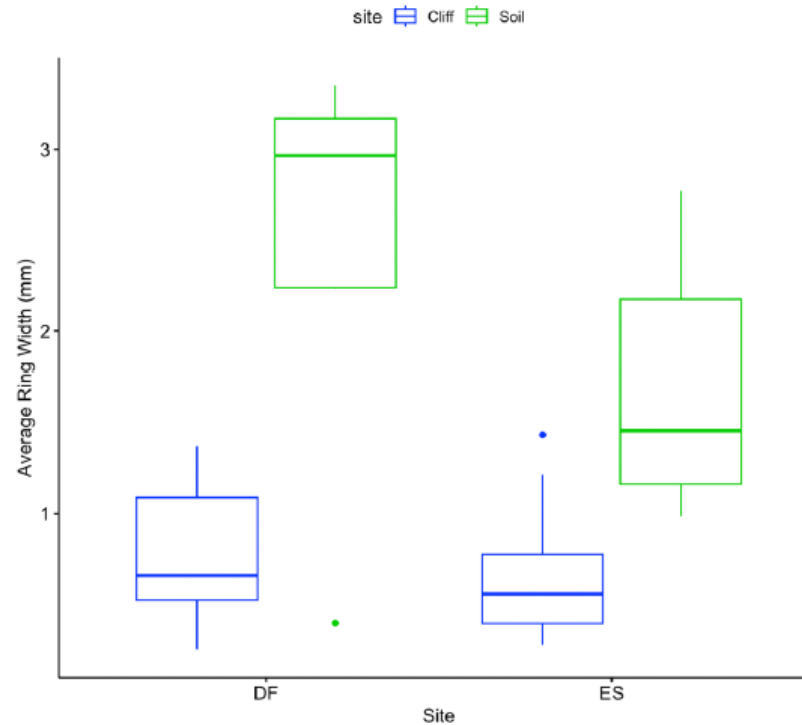


Figure 15: Average ring width (mm) of Douglas Fir (DF) and Engelmann Spruce (ES) trees growing at two sites at Wasootch: a cliff site and glacial till/soil site, measured from tree cores at the root shoot boundary. Douglas Fir is on the left; Spruce is on the right. The Cliff site is in blue, the glacial till site is in green. The samples sizes are as follows: Cliff Douglas Fir ($n=16$), Cliff Engelmann Spruce ($n=29$), Soil Douglas Fir ($n=10$) and Soil Engelmann Spruce ($n=10$).

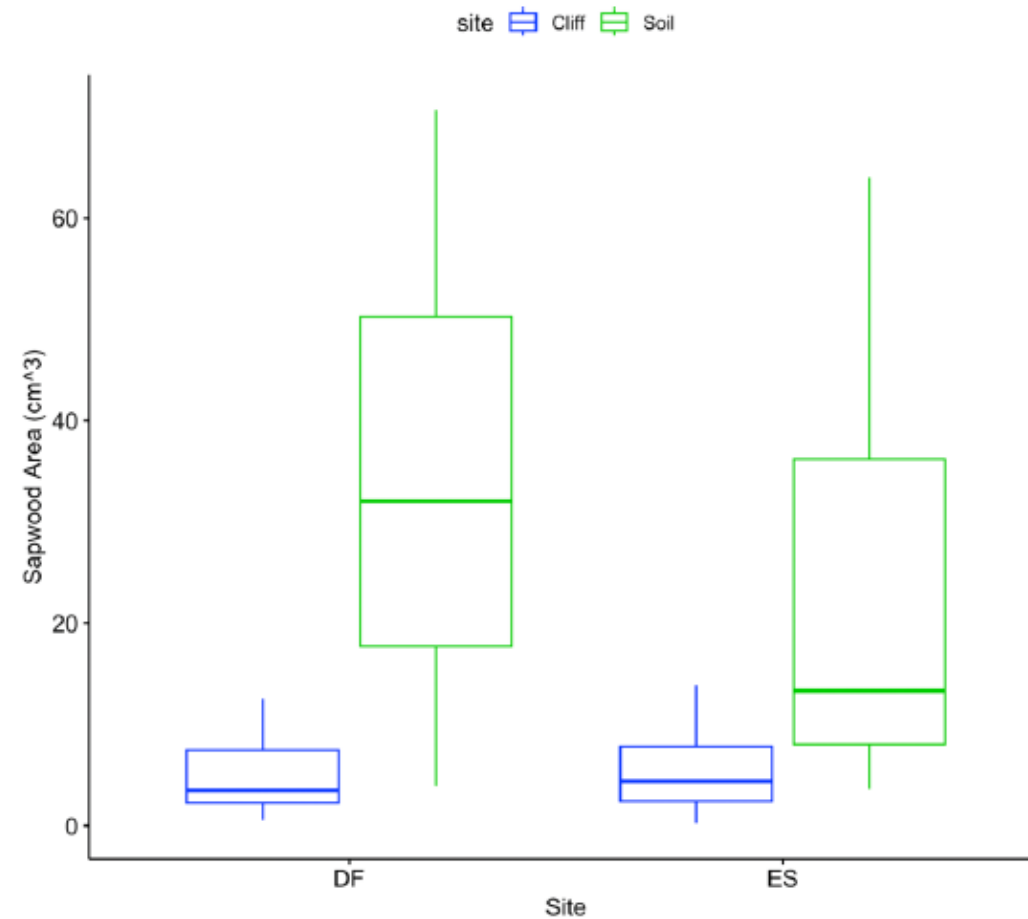


Figure 16: Cross-sectional sapwood area (cm³) of Douglas Fir (DF) and Engelmann Spruce (ES) trees growing at two sites at Wasootch: a cliff site and glacial till/soil site measured from tree cores at the root shoot boundary. Douglas is on the left; Spruce is on the right. The Cliff site is blue, the glacial till site is green. The samples sizes are as follows: Cliff Douglas Fir (n=16), Cliff Engelmann Spruce (n=29), Soil Douglas Fir (n=10) and Soil Engelmann Spruce (n=10).

Tree Cell Characteristics

In 2018, Soil Engelmann Spruce had the largest average earlywood tracheid cell lumen diameter at 28.93 ± 0.65 (mean $\pm$ SE) μm , followed by Soil Douglas Fir at 27.94 ± 0.67 μm and Cliff Spruce at 21.32 ± 0.59 μm . Cliff Douglas Fir had the smallest average earlywood tracheid diameter at 20.82 ± 0.59 μm .

In 2018, Soil Engelmann Spruce had the largest average latewood tracheid cell lumen diameter at 8.90 ± 0.91 (mean $\pm$ SE) μm , followed by Soil Douglas Fir at 7.93 ± 0.44 μm , Cliff Douglas Fir at 7.04 ± 0.57 μm . Cliff Engelmann Spruce had the smallest average latewood tracheid diameter at 6.45 ± 0.59 μm .

In 2018, the Cliff Douglas Fir (3.78 ± 0.13 μm) and Soil Douglas Fir (3.66 ± 0.13 μm) had larger average cell wall thickness than the Cliff Engelmann Spruce (2.54 ± 0.11 μm) and the Soil Engelmann Spruce (2.78 ± 0.14 μm) (Figure 19).

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In all 3 years (2018, 2019 and 2020), the Soil Engelmann Spruce had the largest average tracheid lumen diameter (23.02 ± 0.88 μm , 23.83 ± 0.91 μm and 21.87 ± 0.85 μm), followed by Soil Douglas Fir (18.02 ± 0.98 μm , 20.52 ± 0.90 μm and 19.33 ± 0.97 μm) and Cliff Engelmann Spruce (15.34 ± 0.67 μm , 15.62 ± 0.66 μm and 16.20 ± 0.73 μm). In all 3 years, Cliff Douglas Fir had the smallest average tracheid lumen diameter (14.31 ± 0.74 μm , 15.76 ± 0.75 μm and 15.06 ± 0.75 μm). This trend is also seen in the standardized tracheidograms (Figure 20).

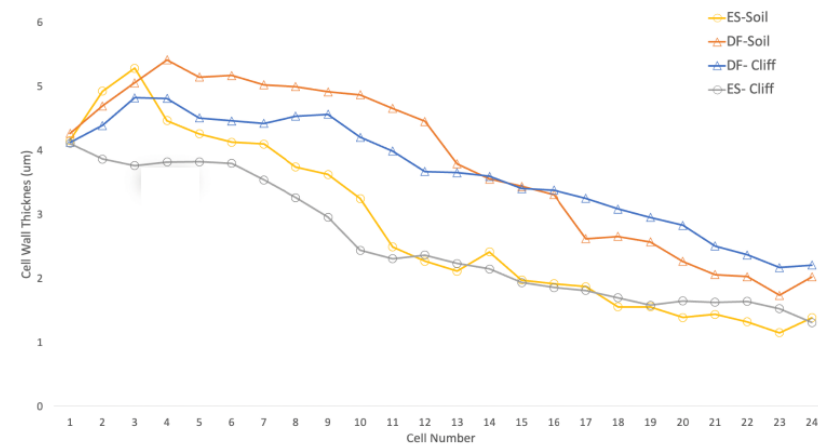


Figure 19: Standardized tracheidograms showing xylem cell wall thickness (μm) of Cliff Douglas Fir, Soil Douglas Fir, Cliff Engelmann Spruce and Soil Engelmann Spruce in 2018. Tracheid were created by standardizing each radial file for each tree to 25 cells using a weighted average group line is an average of 5 trees.

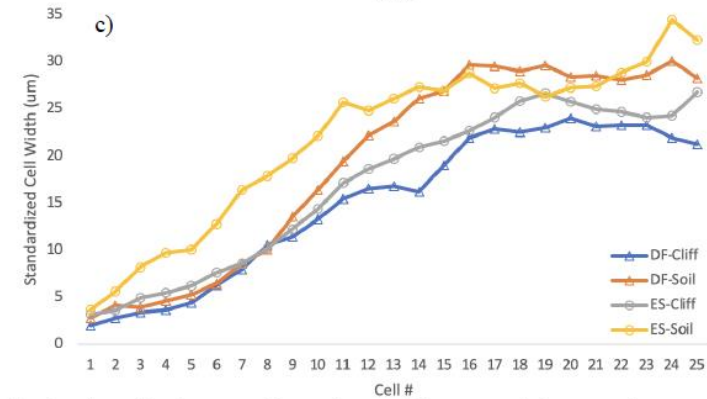
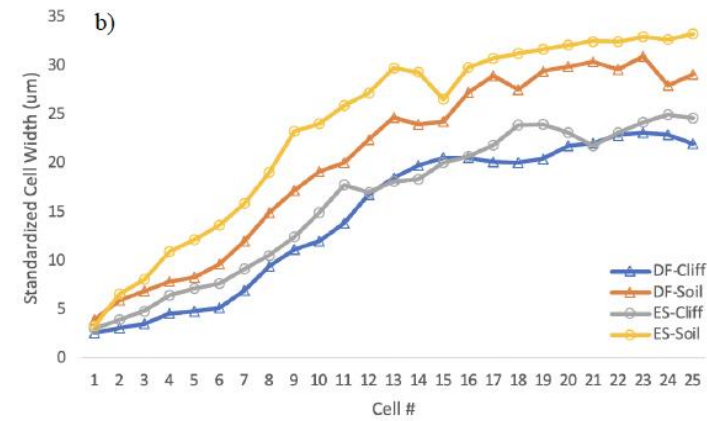
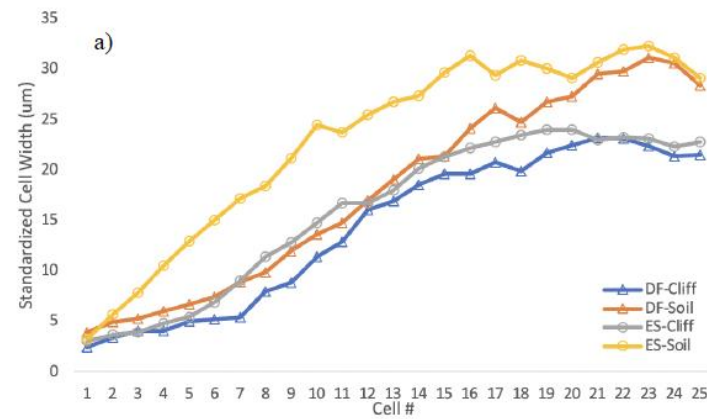


Figure 20: Standardized tracheidograms from a) 2018, b) 2019 and c) 2020, showing xylem cell diameter (μm) of Cliff Douglas Fir, Soil Douglas Fir, Cliff Engelmann Spruce and Soil Engelmann Spruce. Tracheidograms were created by standardizing each radial file for each tree to 25 cells using a weighted average. Each group line is an average of 5 trees.